ByteCover recreation vs original paper

Du et al., ByteCover: Cover Song Identification via Multi-Loss Training (arXiv:2010.14022v2) compared with a small-scale recreation

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Setup

Item	ByteCover paper (arXiv:2010.14022v2)	Paper source	Andy's recreation	Recreation source
Backbone	ResNet-IBN 50 + GeM + BNNeck	p.2 Fig.1, §2.1; p.3 §2.2	ResNet-50-IBN (IBN in stages 1-3) + GeM + BNNeck	model.py, loss.py
Loss	CE(Softmax(W f_c)) + triplet on f_t	p.3 Eq.(2)-(3)	A: CE on f_c only; B: CE + batch-hard triplet on f_t	loss.py
Triplet margin	0.3	p.3 §2.2	0.3 (Euclidean, batch-hard)	loss.py
Optimiser	Adam	p.3 §2.2	Adam, weight decay 0, no schedule	train.py
Learning rate	0.0004	p.3 §2.2	0.0004	train.py
Batch size	32	p.3 §2.2	30 clips (10 works x 3)	train.py
Epochs	not reported	-	100, no early stopping	runs.csv, train.py
Optimizer steps	not reported	-	500 / 1,500 / 3,000 (50/150/300 songs)	PLAN.md
Embedding dim	2048 (128/256 via FC head also tested)	p.3 §2.1; p.4 Table 2	2048	loss.py
CQT implementation	not reported	-	nnAudio CQT2010v2	cache_cqt.py
CQT sample rate	22,050 Hz	p.2 §2.1	22,050 Hz	cache_cqt.py
CQT bins	12 / octave; 84 total	p.2 §2.1	12 / octave; 84 total	cache_cqt.py
CQT window / hop	Hann / 512	p.2 §2.1	Hann / 512	cache_cqt.py
CQT time averaging	factor 100	p.2 §2.1	factor 100 (matches paper)†	cached CQT 84x86 per 200 s
Audio length used	not reported	-	first 200 s (padded to 86 frames)	README.md, dataset.py
Tempo augmentation	not reported	-	none (CQTs cached)	README.md

† Confirmed from the cached CQT arrays (84 x 86 per 200 s clip). The --compress default of 20 in scripts/cache_cqt.py is stale; the runs used --compress 100.

Data

Item	ByteCover paper (arXiv:2010.14022v2)	Paper source	Andy's recreation	Recreation source
Training source	SHS100K training subset	p.3-4 §3.1	SHS-100K-2025 works (YouTube audio)	README.md
SHS100K total (all splits)	8,858 songs / 108,523 rec.	p.3 §3.1	-	-
Split ratio	train:val:test = 8:1:1	p.3 §3.1	-	-
Training songs	not reported (train-split count)	-	50 / 150 / 300 (nested prefixes)	dataset.py
Training recordings	not reported (train-split count)	-	248-250 / 748-750 / 1,498	train_manifest.csv
Recordings per song	not reported	-	5 (299 works), 3 (1 work)	train_manifest.csv
Covers80 leakage removal	not reported	-	13 works excluded	exclude_works.txt
Test set	Covers80: 80 songs x 2 = 160 rec., all pairs	p.4 §3.1	Covers80, 160 recordings	README.md
Similarity	cosine distance	p.4 §3.1	cosine on f_c	loss.py

Train-split size of SHS100K is not reported: the paper gives only 8,858 songs / 108,523 recordings in total, split 8:1:1. Recordings at 50/150 songs are a range: 299 works have 5 recordings, 1 has 3, and data/work_order.txt (not in the repo bundle) decides which prefix holds it.

Compute

Item	ByteCover paper (arXiv:2010.14022v2)	Paper source	Andy's recreation	Recreation source
Hardware	not reported	-	Google Colab, NVIDIA L4 (all 30 runs)	runs.csv, train_loss.csv
Epoch time	not reported	-	0.18 min/epoch (one timing run, size 300)	PLAN.md
Per-run timing log	not reported	-	not logged (no timing column in runs.csv / train_loss.csv)	repo
Min per run, arm A (50/150/300), approx.*	not reported	-	1.1 / 1.2 / 2.4	Drive checkpoint save times*
Min per run, arm B (50/150/300), approx.*	not reported	-	1.6 / 2.5 / 4.9	Drive checkpoint save times*
Elapsed between saves, 30 runs, approx.*	not reported	-	~68 min (A 23, B 45)	Drive checkpoint save times*
Colab compute units used	not reported	-	TBD (from Colab logs)	not found in repo or Drive
Cost	not reported	-	Colab Pro USD 9.99 + 0.80 tax / month (subscription)	Drive: Spending Tracker
Training runs	not reported	-	30 (2 arms x 3 sizes x 5 seeds)	runs.csv

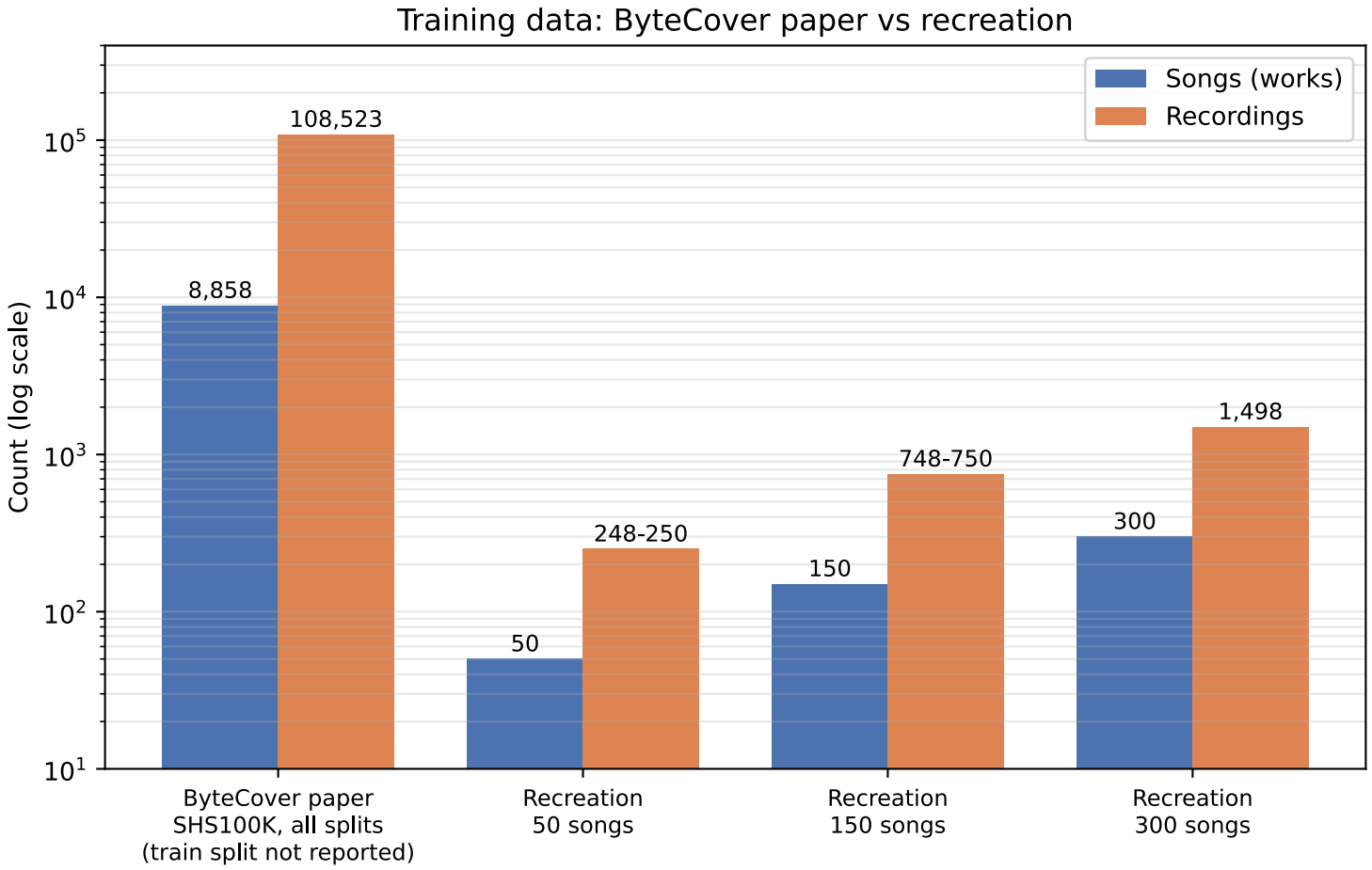
* Approx., from Drive checkpoint save times; the Colab session timed out and resumed, so these are not true wall-clock. Gaps include Covers80 evaluation and saving; the first run of each arm is timed from the results folder / runs.csv creation time.

Results

Item	ByteCover paper (arXiv:2010.14022v2)	Paper source	Andy's recreation	Recreation source
Covers80 mAP	0.906	p.4 Table 1	0.033-0.079 (see per-size table)	runs.csv
Covers80 P@10	0.093	p.4 Table 1	not in provided data	-
Covers80 MR1	3.54	p.4 Table 1	not in provided data	-
Chance mAP (Covers80)	not reported	-	~0.036	brief

Covers80 mAP by training size (n=5 seeds, df=4)

Songs	Recordings	Arm A mAP [95% t-CI]	Arm B mAP [95% t-CI]	B - A paired [95% t-CI]	Paper mAP
50	248-250	0.033 [0.030, 0.036]	0.048 [0.041, 0.055]	0.015 [0.007, 0.023]	0.906
150	748-750	0.045 [0.037, 0.053]	0.056 [0.040, 0.072]	0.011 [-0.012, 0.034]	0.906
300	1,498	0.056 [0.034, 0.079]	0.079 [0.063, 0.096]	0.023 [0.002, 0.045]	0.906



Paper bars: full SHS100K as stated (p.3 §3.1); split 8:1:1, train-split counts not reported.

Figure 1. Training data scale. Songs and recordings on a log scale: the ByteCover paper's SHS100K totals across all splits (8,858 songs / 108,523 recordings; train-split counts not reported) next to Andy's 50, 150 and 300-song training sets.

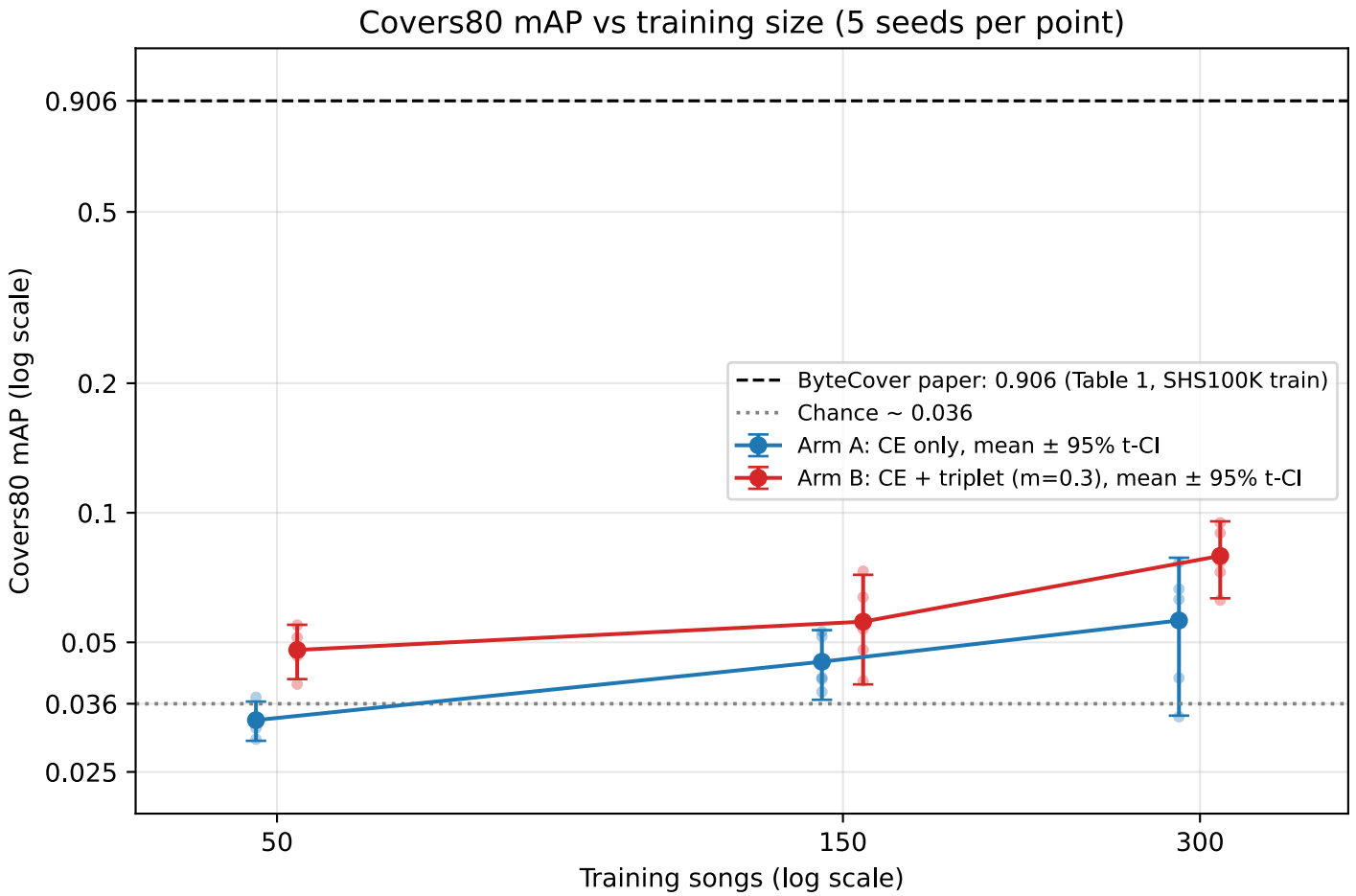


Figure 2. Covers80 mAP vs training size. Mean over 5 seeds with 95% t-intervals (df = 4) for arm A (CE only) and arm B (CE + triplet, margin 0.3), with individual seeds as faint points; dashed line is the paper's Covers80 mAP of 0.906 (Table 1), dotted line is chance (~0.036).

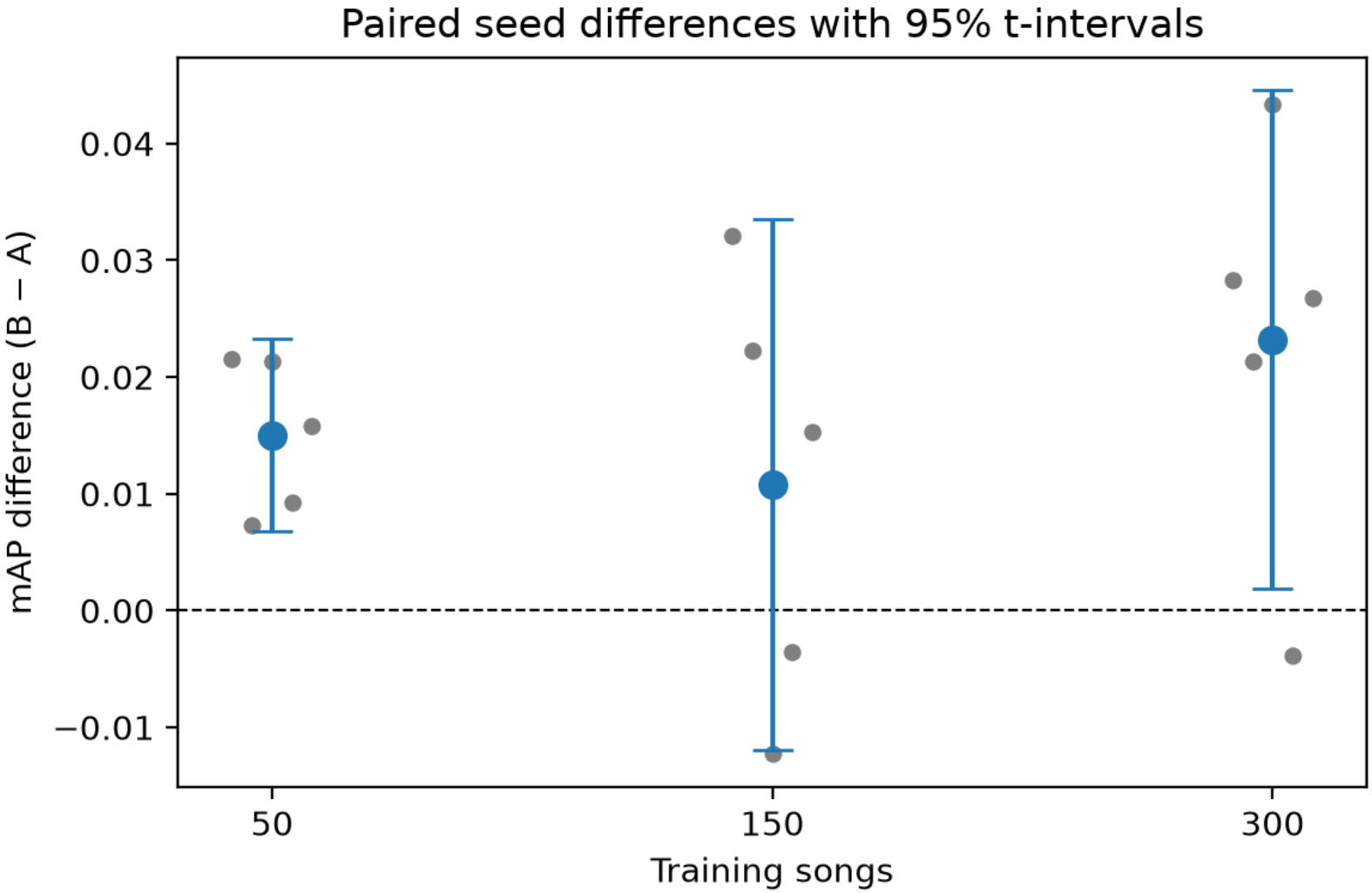


Figure 3. Paired differences in Covers80 mAP (arm B minus arm A) for each seed, with mean and 95% t-interval per training size. The interval excludes zero at 50 and 300 songs and includes zero at 150 songs.

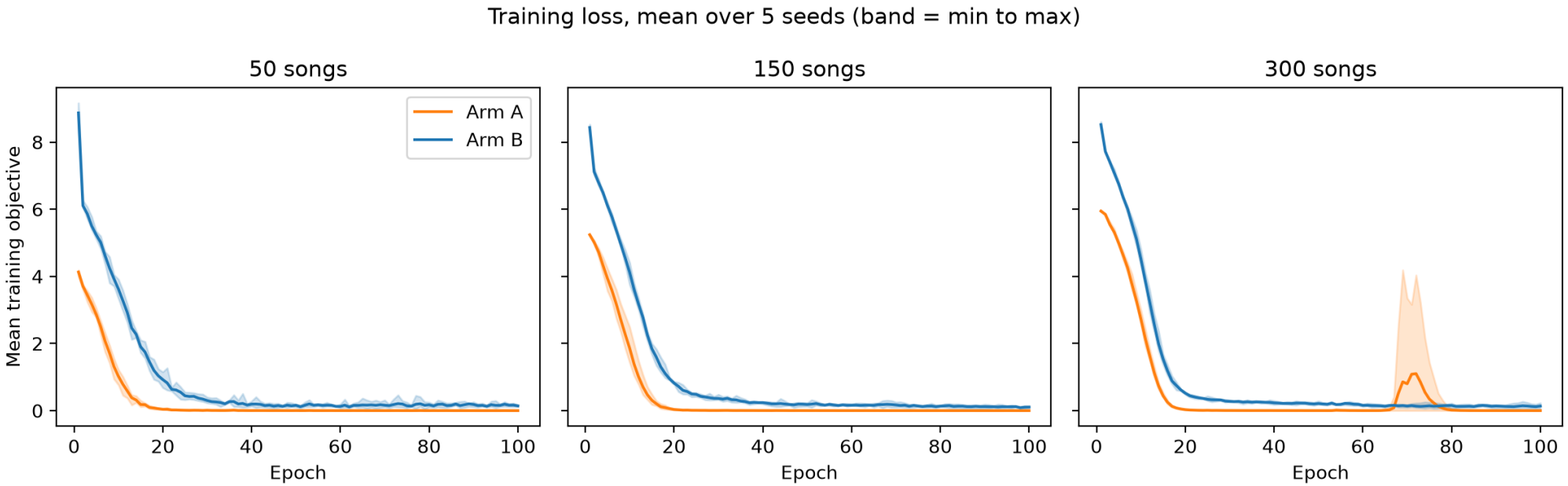


Figure 4. Mean training objective per epoch over 5 seeds (band = min to max) for each arm and training size, 100 epochs each. At 300 songs, arm A shows a temporary rise between roughly epochs 65 and 80 in some seeds before returning near zero.